

VoltStream RAG Quality Evaluation Report

Generated on: 2026-06-19 10:19:15 | Predefined QA Suite

Summary Metrics:

Total Questions: 10 Passed: 10 Failed: 0
Pass Rate: 100.0%

Question 1: What is the most cost-effective efficiency upgrade to reduce heating and cooling energy loss?

Answer: Using caulk and weatherstripping around windows and doors is the most cost-effective efficiency upgrade to seal gaps, prevent drafts, and lower overall heating and cooling energy loss.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 2: By how much can setting the thermostat to 18 degrees C during winter nights reduce annual heating costs?

Answer: Setting your thermostat to 18 degrees C (64 degrees F) during winter nights or while away can reduce annual heating costs by nearly 10%.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 3: What are the main billing components of an electricity bill?

Answer: The main billing components of an electricity bill consist of fixed connection charges, energy consumption charges (measured in kWh), and demand charges. Fixed charges cover grid maintenance, volumetric consumption charges cover energy used, and demand charges cover peak capacity requirements. Additionally, environmental franchise fees and grid taxes also influence the final cost.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 4: When is the best time to run heavy loads to maximize self-consumption with solar offsets?

Answer: Setting up smart schedules to run heavy loads when solar generation is highest, typically between 11 AM and 2 PM, maximizes self-consumption and solar offsets.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 5: What are the typical peak electricity demand hours during summer and winter months?

Answer: The typical peak electricity demand hours during summer and winter months are between 7 PM and 9 PM. During these times, the grid experiences its highest load.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 6: How can you significantly reduce billing costs related to heavy appliance usage?

Answer: Consumers can significantly reduce billing costs related to heavy appliance usage by shifting heavy appliance usage (e.g., washing machines, dishwashers, and water heaters) to off-peak hours (after 10 PM or before 6 AM).

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

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Question 7: How frequently do smart meters transmit usage data to the utility provider?

Answer: Smart meters transmit usage data every 15 minutes to the utility provider. This high-frequency logging enables utility providers to offer dynamic pricing.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 4/5 Status: PASS

Question 8: How much of residential energy usage is typically accounted for by phantom or standby load?

Answer: Standby load accounts for up to 10% of residential energy usage.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 9: What is the recommended panel angle for peak annual performance of a residential solar array?

Answer: For peak annual performance of a residential solar array, the recommended panel angle should be equal to the local latitude, generally between 30 degrees and 45 degrees.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 10: How much does cleaning solar panels every 3 months increase overall output efficiency compared to uncleaned panels?

Answer: Cleaning solar panels every 3 months increases overall output efficiency by 15% compared to uncleaned panels.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS